

Engineering Statistics

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Chapter 9: Inference for Proportions

Outline

- ▶ One-sample inference for a population proportion
- ▶ Confidence intervals and adjusted intervals
- ▶ Hypothesis testing for one proportion
- ▶ Inference for the difference of two proportions
- ▶ Hypothesis tests with pooled proportions

Assumptions for inference about p

1. We have a good **simple random sample** (SRS) from the population of interest.
2. The population is **Bernoulli distribution**, $Ber(p)$.
3. **We don't know p .**
4. We have a large enough sample of size n .

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} Ber(p)$$

Assumptions for inference about p

For n large enough, $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$,

$$\hat{P} \longrightarrow N\left(p, \frac{p(1-p)}{n}\right),$$

$$T = \frac{\hat{P} - p}{\sqrt{\frac{p(1-p)}{n}}} \sim N(0, 1)$$

by the central limit theorem.

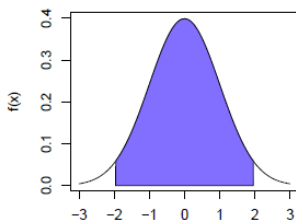
CI for a population proportion p

Since **we don't know** p , we use **standard error**

$$\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

Then, for n large enough ($n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$), a **level** $(1 - \alpha)100\%$ **confidence interval** for the population proportion p :

$$\hat{p} \pm z_{1-\frac{\alpha}{2}} \times \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$



Why not t ?

- ▶ t -distribution needs **independence** between numerator and denominator of a t -statistic.
 - ▶ e.g. $\bar{X}$ and S^2 are independent from Normally distributed case.
 - ▶ The standard error $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ depends on $\hat{p}$
 - ▶ the numerator and denominator are NOT independent.
- ▶ For large n , we can use $z_{1-\frac{\alpha}{2}}$ by
 - ▶ the central limit theorem
 - ▶ other theoretical backgrounds
 - ▶ e.g. Slutsky's theorem, continuous mapping theorem, ...

Example: Prevalence of HPV infections

- ▶ A national survey found that 515 of a sample of 1921 women aged 14 to 59 years tested positive for HPV.
- ▶ Assumption:

$$X_1, X_2, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$$

with large $n = 1921$. If tested positive, $X_i = 1$.

Example: Prevalence of HPV infections

- ▶ $\hat{p} = 515/1921 = 0.2681$.
- ▶ $\sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.0101$
- ▶ With confidence level 95%, $z_{1-\alpha/2} = 1.96$

A 95% **confidence interval** for the proportion difference:

$$0.2681 \pm 1.96 \times 0.0101 \iff (0.2483, 0.2879).$$

More accurate CI for p

Unfortunately, the actual confidence levels from

$$\hat{p} \pm z_{1-\frac{\alpha}{2}} \times \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

are often quite inaccurate unless the sample is very large.

- ▶ From simulations, the actual confidence level is usually less than what we set.
- ▶ Also, accuracy does not get better as n increases.

More accurate CI for p

- ▶ **Plus four CI for p**

- ▶ A simple modification can improve the accuracy.
- ▶ Add four imaginary observations: **two successes, two failures.**
- ▶ **Plus four estimate of p :**

$$\tilde{p} = \frac{\#success + 2}{n + 4}$$

- ▶ Use this when $(1 - \alpha) \geq 90\%$ and $n \geq 10$.
- ▶ It is recommended to always use this method for CI.

Example: Prevalence of HPV infections

- ▶ $\hat{P} = 515/1921 = 0.2681$.
- ▶ $\tilde{p} = (515 + 2)/(1921 + 4) = 0.2686$.
- ▶ $\sqrt{\frac{\tilde{p}(1-\tilde{p})}{n+4}} = 0.0101$
- ▶ With confidence level 95%, $z_{1-\alpha/2} = 1.96$

A 95% **confidence interval** for the proportion difference:

$$0.2686 \pm 1.96 \times 0.0101 \iff (0.2488, 0.2884).$$

Hypothesis tests for p

- ▶ A hypothesis test for a population proportion p
 - ▶ To test $H_0 : p = p_0$.

$$\hat{P} \stackrel{H_0}{\sim} N\left(p_0, \frac{p_0(1-p_0)}{n}\right).$$

- ▶ Then, the test statistic is

$$T = \frac{\hat{P} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} \stackrel{H_0}{\sim} N(0, 1),$$

when $np_0 \geq 10$ and $n(1-p_0) \geq 10$.

Example: Soda consumption

- ▶ In 2015, a survey reported that 615 of a random sample of 1009 American adults said they try to avoid drinking soda.
- ▶ $H_0 : p = 0.5$ vs $H_a : p > 0.5$
- ▶ Under $H_0 : p = 0.5$ is true,

$$\frac{\hat{P} - 0.5}{\sqrt{\frac{0.5(1-0.5)}{1009}}} \stackrel{H_0}{\sim} N(0, 1).$$

- ▶ $\hat{p} = 0.6095$.
- ▶ The observed $t = 6.96$.
- ▶ P-value = $P(T > 6.96) \approx 0$.

Standard errors: CI vs Hypothesis testing

- ▶ CI: using observed sample proportion,

$$\sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}.$$

- ▶ Hypothesis testing: using the null value,

$$\sqrt{\frac{p_0(1 - p_0)}{n}}.$$

Using AI to Learn: One Population Proportion

Ask AI

Explain the difference between the standard errors used for a confidence interval and a hypothesis test for a population proportion. Why do we use p_0 in the hypothesis test?

Verify the Explanation

Check the assumptions, identify the null hypothesis, and explain why the standard error depends on the purpose of the inference.

Assumptions for inference of two proportions

1. We have a good **simple random samples** (SRS) from the populations of interest.
2. The populations are **Binomial distribution**: $Ber(p_1)$ and $Ber(p_2)$.
3. **We don't know** p_1, p_2 .
4. We have a large enough sample of size n_1 and n_2 .

$$X_{11}, X_{12}, \dots, X_{1n_1} \stackrel{iid}{\sim} Ber(p_1),$$

$$X_{21}, X_{22}, \dots, X_{2n_2} \stackrel{iid}{\sim} Ber(p_2).$$

Notations

Population	Sample size	Population proportion	Sample proportion
1	n_1	p_1	$\hat{p}_1$
2	n_2	p_2	$\hat{p}_2$

We want to compare the two population proportions, by giving a confidence interval for their difference ($p_1 - p_2$) or by the hypothesis testing of no difference. We will use $\hat{p}_1 - \hat{p}_2$.

Sampling distributions for the difference of proportions

Two sample are **independent**. When n_1 and n_2 are large:

$$(\hat{P}_1 - \hat{P}_2) \sim N\left(p_1 - p_2, \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right).$$

Then,

$$\frac{(\hat{P}_1 - \hat{P}_2) - (p_1 - p_2)}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \sim N(0, 1)$$

Confidence intervals for $p_1 - p_2$

A level $(1 - \alpha)100\%$ **confidence interval** for $p_1 - p_2$:

$$\hat{p}_1 - \hat{p}_2 \pm z_{(1-\frac{\alpha}{2})} \times \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}.$$

Example

A foreign survey was conducted to assess the perception of their government performance among males and females. Among a sample of 1000 individuals, 516 were males and 484 were females. Of the male respondents, 31.8% gave a positive evaluation of presidential performance, while 43.6% of female respondents evaluated positive.

$$X_{11}, X_{12}, \dots, X_{1n_1} \stackrel{iid}{\sim} \text{Ber}(p_1),$$

$$X_{21}, X_{22}, \dots, X_{2n_2} \stackrel{iid}{\sim} \text{Ber}(p_2).$$

with $n_1 = 516$ and $n_2 = 484$. If evaluated positive, $X = 1$.

Example

- ▶ $\hat{p}_1 = 164/516 = 0.318$, $\hat{p}_2 = 211/484 = 0.436$.
- ▶ $\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}} = 0.0305$
- ▶ With confidence level 95%, $z_{1-\alpha/2} = 1.96$

A 95% **confidence interval** for the proportion difference:

$$0.318 - 0.435 \pm 1.96 \times 0.0305 \iff (-0.1768, -0.0572).$$

More accurate CI for $p_1 - p_2$

- ▶ **Plus four CI for $p_1 - p_2$**

- ▶ A simple modification can improve the accuracy.
- ▶ Add **four** imaginary observations:
 - ▶ **one success, one failure** for each of the two samples.
- ▶ **Plus four estimates** of p_1 and p_2 :
 - ▶

$$\tilde{p}_1 = \frac{\#success + 1}{n_1 + 2}, \quad \tilde{p}_2 = \frac{\#success + 1}{n_2 + 2}$$

- ▶ Use this when $n_1 \geq 5$ and $n_2 \geq 5$.

Example

- ▶ $\tilde{p}_1 = (164 + 1)/(516 + 2) = 0.322$,
 $\tilde{p}_2 = (211 + 1)/(484 + 2) = 0.436$.
- ▶ $\sqrt{\frac{\tilde{p}_1(1-\tilde{p}_1)}{n_1+2} + \frac{\tilde{p}_2(1-\tilde{p}_2)}{n_2+2}} = 0.0305$
- ▶ With confidence level 95%, $z_{1-\alpha/2} = 1.96$

A 95% **confidence interval** for the proportion difference:

$$0.322 - 0.436 \pm 1.96 \times 0.0305 \iff (-0.1738, -0.0542).$$

Hypothesis tests for the difference of proportions

- ▶ We want to test whether there is any difference between two population proportions
 - ▶ $H_0 : p_1 - p_2 = 0$.
 - ▶ Under H_0 , let $p = p_1 = p_2$.
- ▶ Under H_0 is true,

$$\hat{P}_1 - \hat{P}_2 \stackrel{H_0}{\sim} N\left(0, p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)\right).$$

Hypothesis tests for the difference of proportions

- For standard error, instead of using separate $\hat{p}_1$ and $\hat{p}_2$, use the **pooled sample proportion**

$$\hat{p} = \frac{\text{\#success in sample1} + \text{\#success in sample2}}{n_1 + n_2}$$

under $H_0 : p_1 = p_2 = p$.

- Then, the test statistic is

$$T = \frac{(\hat{P}_1 - \hat{P}_2) - 0}{\sqrt{\hat{p}(1 - \hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \xrightarrow{H_0} N(0, 1).$$

Use this test when the number of successes and failures are each 5 or more in both samples.

Example: HIV vaccine

- ▶ A study conducted in Thailand, enrolled HIV-negative men and women between the ages of 18 and 30 years who were at an average risk of infection.
- ▶ Study design:
 - ▶ Randomized and double blinded
 - ▶ Placebo group: $n_1 = 8198$, Vaccine group: $n_2 = 8197$.
 - ▶ All participants received counseling on how to avoid HIV infection, and regularly tested for HIV infection over a three-year period.
- ▶ $H_0 : p_1 = p_2$ vs $H_a : p_1 > p_2$.

Example: HIV vaccine

- ▶ $\hat{p}_1 = 74/8198 = 0.0090$, $\hat{p}_2 = 51/8197 = 0.0062$.
- ▶ Pooled sample proportion:
 $\hat{p} = (74 + 51)/(8198 + 8197) = 0.0076$.
- ▶ The observed test statistic: $t = 2.064$
- ▶ $P(T > 2.064) = 0.0195$.

Self-Study: Inference for Two Proportions

The Problem

Suppose we have two independent populations with proportions p_1 and p_2 . How can we estimate the difference between their population proportions?

Your AI Learning Task

Ask AI to explain:

- ▶ The assumptions for comparing two population proportions.
- ▶ The sampling distribution of $\hat{p}_1 - \hat{p}_2$.
- ▶ How to construct and interpret a confidence interval for $p_1 - p_2$.

Challenge: What does it mean if the confidence interval contains zero?

Self-Study: Inference for Two Proportions

The Problem

How can we test whether two independent populations have the same proportion?

$$H_0 : p_1 = p_2 \quad \text{vs.} \quad H_a : p_1 \neq p_2$$

Your AI Learning Task

Ask AI to explain:

- ▶ How to construct the test statistic.
- ▶ Why a pooled proportion is used under H_0 .
- ▶ How to calculate and interpret the P-value.

Challenge: Why use pooling for testing but not for the usual confidence interval?

Chapter Summary

- ▶ **Use sample proportions to learn about population proportions.** Sampling distributions quantify uncertainty in estimates.
- ▶ **Construct confidence intervals to estimate plausible values.** Check sample-size conditions and consider adjusted intervals.
- ▶ **Use hypothesis tests to evaluate claims about proportions.** Compute P-values under the null hypothesis.
- ▶ **Distinguish one- and two-population procedures.** For testing equality of two proportions, use a pooled estimate under the null.
- ▶ **Interpret statistical evidence carefully.** Verify assumptions, calculations, and conclusions, including AI-generated explanations.